MIL-P-50603B(AR)
1 July 1986
SUPERSEDING
MIL-P-50603A(MU)
5 June 1978

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M115

This Specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the metal parts and the assembly of one type of percussion primer, M115, for use in 20mm and 25mm cartridges.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DoDISS) and supplement thereto cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168	-	Ammunition Lot Numbering

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1305

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a patrt of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on date of solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT & ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9387196	-	Packing and Marking; Box, Wirebound; Two Boxes, M2A1, for Primers
10542585	-	Primer, Percussion, M115
10542655	-	Cartridge, 20mm Target Practice, Tracer, M206A1
12013224	-	Cartridge, 25mm, TP-T, M793

INSPECTION EQUIPMENT DRAWINGS

7553194	-	Ball, 20mm
8648468	-	Breech Block Assembly
8648625	-	Fixture, Sensitivity Testing, Percussion Primers
11075138	-	Indicator Gage

PUBLICATIONS

ORD-SIP-S314	-	Visual Inspection Standards for Small Arms Ammunition Primer Defects
AMCR 715-505 Volume 8	-	Ammunition Ballistic Acceptance Test Methods - Test Procedures for 20mm Cartridges
AS12013566	-	25mm Ammunition Ballistic Test Method

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Primer. The primer shall comply with all requirements specified on drawing (dwg) 10542585, all associated drawings and with all requirements specified in applicable specifications. It shall be packed in accordance with Dwg. 9387196.

3.3 Sensitivity. The mean critical height ($\overline{H}$) plus five standard deviations (5s) for the sample primers shall not exceed 19 inches. In addition, the mean critical height ($\overline{H}$) minus two standard deviations (2s) for the sample primers shall not be less than 2 inches.

3.4 Action Time. The action time for individual sample primers and the average action time for the sample primers plus four standard deviations (4s), when assembled into 20mm cartridges and tested at ambient temperature, shall not exceed four milliseconds. The action time for individual sample primers and the average action time for the sample of primers plus four standard deviations, (4s), when assembled into 25mm cartridges and tested at -54°C shall not exceed six milliseconds.

3.5 Function and casualty. The sample primers shall function without casualty when assembled into test cartridges and fired in the applicable weapon.

3.6 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval (see 6.6).

3.7 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3 and 6.2.d).

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3.8 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough and workmanlike manner and shall be free of oil, grease and other foreign material. In addition, the primer shall comply with the standards specified in ORD-SIP-S314.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. In addition, unless specified herein or in the contract the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawing	Quantity
Cup	10542586	5/Tool
Anvil	10542589	5/Tool
Primer, Percussion, M115	10542585	2000
Primer, Percussion, M115 (prior to insertion of anvil)	10542585	800

4.3.2 Inspections to be performed. First article assemblies and components shall be subjected to any or all of the examinations and tests specified in Table I and to any or all requirements of the applicable drawings.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50603B(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2			DRAWING NUMBER
	Primer Percussion M115 and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup (Dwg. 10542586)				
	Examination for Defects	5/ tool		3.2	1/
	Anvil (Dwg. 10542589)				
	Examination for Defects	5/ tool		3.2	1/
	Primer Percussion M115 (Prior to Insertion of Anvil) (Dwg. 10542585)				
	Examination for Defects	800		3.2	4.4.2.2
	Primer Percussion, M115 (Dwg. 10542585)				
	Examination for Defects	2000		3.2	4.4.2.3
	Sensitivity	2/3/	4/	3.3	4.5.2, Dwgs 7553194, 8648468 5/ & 8648625

NOTES: 1/ The components shall be inspected with standard measuring and test equipment for all drawing requirements.
2/ One hundred primers shall be tested at each height.
3/ The primers for these tests shall be taken from an acceptable "Examination for Defects" quantity.
4/ If the average drop height ($\bar{H}$) plus five standard deviations (5S) exceeds the maximum requirement or if $\bar{H}$ minus 2S is less than the minimum requirement the first article sample shall be rejected.
5/ Dwg. 8648468 to be used in lieu of Dwg. 7553215 which is shown on Dwg. 8648625.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-P-50603B(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER See Below
	Primer Percussion M115 and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Action Time	100 3/	6/ or 7/	3.4	4.5.3.1 or 4.5.3.2 8/
	Function and Casualty	400 3/	9/	3.5	4.5.4.1 or 4.5.4.2 8/

NOTES: 6/ If the sample average action time for 20mm cartridges plus four standard deviations exceed four milliseconds or an individual primer action time of the test sample exceeds four milliseconds, the first article quantity shall be rejected.
7/ If the sample average action time for 25mm ammunition plus four standard deviations exceed six milliseconds or an individual primer action time of the test sample exceeds six milliseconds, the first article quantity shall be rejected.
8/ The primers shall be tested in the cartridge for which they are being procured.
9/ If one hangfire (see 6.7.1), two or more misfires, two or more pierced primers, or ten or more primer leaks occur the first article quantity shall be rejected.

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4.3.3 Rejection. The First Article quantity shall be rejected upon failure of any assembly or component in the sample to comply with any visual or dimensional characteristics or upon failure to comply with the criteria for any test. The Government reserves the right to terminate its inspection whenever a component or assembly fails to comply with any visual or dimensional characteristic or components or assemblies fail to comply with the criteria for any test.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. The inspection lot shall comply with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirments. Lot numbering, as required, shall be in accordance with MIL-STD-1168. A lot shall consist of:

a. Primers assembled by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Parts produced by one manufacturer, in one unchanged process, in accordance with the same drawing and drawing revisions, and same specification and specification revision.

4.4.2 Examination. Unless otherwise specified in the classification of defects and test tables, sampling plans for the individual major and minor defects listed shall be in accordance with MIL-STD-105, Inspection Level II. Alternatively, inspection, except for inspection of packaging, packing and marking, of major or minor defects as listed herein may be considered on a class basis using an AQL of 1.5% for major and 2.5% for minor defects. However, where major defects are considered on a class basis, the acceptance number for any individual defect shall be limited by an AQL of 0.40%. One hundred percent examination shall be performed for all critical defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50603B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 10542586
4.4.2.1	Cup				NEXT HIGHER ASSEMBLY 10542585
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Height, Maximum		100%	3.2	Gage
MAJOR	None defined.				
MINOR	None defined.				

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Primer, Percussion, M115 (Prior to Insertion of Anvil)				10542585
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL 1	Height of Composition, Minimum		100%	3.2	Automatic Inspection Equipment, 4.5.1
MAJOR	None defined.				
MINOR	None defined.				

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50603B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER
4.4.2.3	Primer Percussion M115				10542585
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	1/				
Critical					
1	Anvil missing (1)		100%	3.2	Visual
2	Presence of more than one anvil (2)		100%	3.2	Visual
3	Pellet missing (3)		100%	3.2	Visual
4	Inverted or sidewise inserted anvil (7)		100%	3.2	Visual
5	Cocked anvil (11)		100%	3.2	Visual
6	Partial, anvil (36)		100%	3.2	Visual
7	Vent missing (37)		100%	3.2	Visual
8	Stained foil (42)		100%	3.2	Visual
Major					
101.	Partial or missing foil (4) (41)		100%	3.2	Visual
102.	Presence of more than one foil (5)		0.25%	3.2	Visual
103.	Slipped foil (6)		0.25%	3.2	Visual
104.	Crushed or mutilated (8)		0.25%	3.2	Visual
105.	Loose composition (9)		0.25%	3.2	Visual
106.	Deep seated anvil (10)		0.25%	3.2	Visual
107.	Low anvil (12)		0.25%	3.2	Visual
108.	High anvil (13)		0.25%	3.2	Visual
109.	Foreign material (17)		0.25%	3.2	Visual
110.	Bell shaped (18)		0.25%	3.2	Visual
111.	Bad bevel (19)		0.25%	3.2	Visual
112.	Brass shavings (20)		0.25%	3.2	Visual
113.	Cracked (21)		0.25%	3.2	Visual
114.	Crooked or eared (22)		0.25%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
4.4.2.3	Primer Percussion M115				10542585
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
115.	Dents (23)		0.25%	3.2	Visual
116.	Crown on flat cup (25)		0.25%	3.2	Visual
117.	Laminated metal (27)		0.25%	3.2	Visual
118.	Ringed (28)		0.25%	3.2	Visual
119.	Scratches (29)		0.25%	3.2	Visual
120.	Nicked (30)		0.25%	3.2	Visual
121.	V or U (31)		0.25%	3.2	Visual
122.	Low cup (32)		0.25%	3.2	Visual
123.	Chipped (34)		0.25%	3.2	Visual
124.	Filled (35)		0.25%	3.2	Visual
125.	Laminated metal (38)		0.25%	3.2	Visual
126.	Heavy (43)		0.25%	3.2	Visual
127.	Shellac missing or excessive		0.25%	3.2	Visual
128.	Loose anvil		0.25%	3.2	Manual 2/
129.	Chamfer missing		0.25%	3.2	Visual
130.	Excess tolerance between long axis and short axis diameters		0.25%	3.2	Dwg. 11075138
131.	Height		0.25%	3.2	Gage
132.	Diameter		0.25%	3.2	Gage
133.	Sensitivity	3/4/	4/	3.3	4.5.2, Dwgs. 7553194, 8648468 5/ and 8648625
134.	Action time	503/	6/or7/	3.4	4.5.3.1 or 4.5.3.2 8/
135.	Function and Casualty	2003/	9/	3.5	4.5.4.1 or 4.5.4.2 8/

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER 10542585
4.4.2.3	Primer Percussion M115				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201.	Stained or corroded metallic components (16)		0.40%	3.2	Visual
202.	Dents (23)		0.40%	3.2	Visual
203.	Laminated metal (27)		0.40%	3.2	Visual
204.	Ringed (28)		0.40%	3.2	Visual
205.	Scratches (29)		0.40%	3.2	Visual
206.	Nicked (30)		0.40%	3.2	Visual
207.	Burred (33)		0.40%	3.2	Visual
208.	Chipped (34)		0.40%	3.2	Visual
209.	Ringed (39)		0.40%	3.2	Visual
210.	Unsymmetrical vent holes (40)		0.40%	3.2	Visual
211.	Evidence of Poor Workmanship		0.40%	3.8	Visual

NOTES:

1/ Numbers within parenthesis represent corresponding defect numbers in ORD-SIP S314.

2/ The anvil shall not fall out when the primer is inverted.

3/ Only primers which have met all visual and dimensional requirements specified in this Table shall be used for this test.

4/ Fifty primers shall be tested at each height. If the mean critical height ($\bar{H}$) plus five standard deviations (5S) exceeds the maximum requirement or if $\bar{H}$ minus 2S is less than the minimum requirement the lot shall be rejected subject to one retest. A retest shall consist of 100 primers tested at each height. If on retest the sample fails to meet the above criteria, the lot shall be rejected.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
4.4.2.3	Primer Percussion M115				10542585
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	5/ Dwg. 8648468 to be used in lieu of Dwg. 7853215 which is shown on Dwg. 8648625.				
	6/ If the sample average action time for 20mm cartridges plus four standard deviations exceed four milliseconds or an individual primer action time of the test sample exceeds four milliseconds, the lot shall be rejected.				
	7/ If the sample average action time for 25mm ammunition plus four standard deviations exceed six milliseconds or an individual primer action time of the test sample exceeds six milliseconds, the lot shall be rejected.				
	8/ The primers shall be tested in the cartridge for which they are being procured.				
	9/ If one hangfire (see 6.7.1), two or more misfires, two or more pierced primers, or 10 or more primer leaks occur the lot shall be rejected. If one misfire or one pierced primer, or four or more primer leaks occur a second sample consisting of double the quantity specified shall be fired. If in the cumulative sample (first test and retest) one hangfire, two or more misfires two or more pierced primers or 10 or more primer leaks occur the primer lot shall be rejected.				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50603B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9387194
4.4.2.4	Carton, Packing, for Primer, M115				NEXT HIGHER ASSEMBLY 9387195
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100% 1/	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major	None defined.				
Minor					
201.	Incorrect number of primers		2.5%	3.2	Visual
202.	Primers improperly placed in tray		2.5%	3.2	Visual
203.	Tray damaged or cracked		2.5%	3.2	Visual
204.	Sleeve damaged or torn		2.5%	3.2	Visual
205.	Tray loose in sleeve		2.5%	3.2	Manual
206.	Marking improper, incorrect or missing		2.5%	3.2	Visual

NOTES: 1/ Sampling plan shall be in accordance with MIL-STD-105, Inspection Level 1.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9387195
4.4.2.5	Packing and Marking; Box, Ammunition, M2A1, for Primer (Plastic Tray)				NEXT HIGHER ASSEMBLY 9387196
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.		1/		
Major					
101.	Ammunition box seal missing, damaged or improperly assembled		100%	3.2	Visual
102.	Bottom filler missing		1.0%	3.2	Visual
103.	Improper number of primer cartons		1.0%	3.2	Visual
104.	Top filler missing		1.0%	3.2	Visual
105.	Ammunition box improperly closed		1.0%	3.2	Visual/Manual
Minor					
201.	Marking on ammunition box improper, incorrect or missing		2.5%	3.2	Visual

NOTES:

1/ Sampling plan shall be in accordance with MIL-STD-105, Inspection Level I.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50603B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Packing and Marking; Box, Wirebound; Two-Boxes, M2A1, for Primer				9387196
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100% 1/	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101.	End filler (2) missing		1.0%	3.2	Visual
102.	Wood separator missing		1.0%	3.2	Visual
103.	Incorrect number of ammunition boxes		1.0%	3.2	Visual
104.	Wirebound box improperly closed		1.0%	3.2	Visual/Manual
Minor					
201.	Marking on wirebound box improper, incorrect or missing		2.5%	3.2	Visual

NOTES: 1/ Sampling plan shall be in accordance with MIL-STD-105, Inspection Level I.

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4.4.3. Reduced testing, function and casualty. When the manufacturer of the primer is also the cartridge loader, the function and casualty test shall be performed on the first article sample only.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" columns in Table I and paragraphs 4.4.2.1 through 4.4.2.3. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.2 herein.

4.5 Methods of inspection.

4.5.1 Height of composition, minimum. The design of the automatic inspection equipment used to inspect for this characteristic shall be submitted for approval in accordance with paragraph 6.2.4 of MIL-A-48078 and paragraph 6.3 of this specification. The correlation analysis between the height of composition, minimum, and the minimum dry weight of primer composition shall be included in this submission.

4.5.2 Sensitivity. The method of test shall be as specified in AMCR 715-505, Volume 8. The complete rundown test shall be used. Test primers shall be at ambient temperature.

4.5.3 Action time.

4.5.3.1 Action time for 20mm. The method of test shall be as specified in AMCR 715-505, Volume 8. The test sample primers shall be assembled into cartridge cases that have been previously accepted. In addition, each cartridge case shall be inspected for depth and diameter of primer pocket and length to shoulder basic diameter. The sample primed cases shall be serialized and the weight of propellant recorded. After assembly as a Cartridge, 20mm, Target Practice, Tracer, M206A1, Dwg. 10542655, each test cartridge shall be inspected for projectile crimp improper or incomplete, depth of primer seating, primer crimp missing, loose primer, maximum profile and alignment, and presence of foreign matter on the primer. The sample test cartridges shall be fired at ambient temperature in a test barrel.

4.5.3.2 Action time for 25mm. The method of test shall be as specified in AS12013566. The test sample primers shall be assembled into cartridge cases that have been previously accepted. In addition, each cartridge case shall be inspected for depth and diameter of primer pocket and length to shoulder basic diameter. The sample primed cases shall be inspected for absence of booster, serialized and the weight of propellant recorded. After assembly as a Cartridge, 25mm, TP-T, M793, Dwg. 12013224 each test cartridge shall be inspected for projectile crimp

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missing or incomplete, depth of primer seating, primer crimp missing, loose primer, maximum profile and alignment, and presence of foreign matter on the primer. The sample test cartridges, conditioned for a minimum of four hours in a conditioning chamber which has been established at -54°C ± 3°C, shall be handled in accordance with paragraph 5.2.3.10 of AS12013566 and fired in a test barrel.

4.5.4 Function and casualty.

4.5.4.1 Function and casualty for primers for use in 20mm cartridges and weapon systems. The method of test shall be as specified in AMCR 715-505, Volume 8, except that witness panels arel not required. The test sample primers shall be assembled into cartridge cases that have been previously accepted. In addition, each cartridge case shall be inspected for depth and diameter of primer pocket and length to shoulder basic diameter. After assembly as a Cartridge, 20mm, Target Practice, Tracer, Dwg. 10542655 M206A1, each test cartridges shall be inspected for projectile crimp improper or incomplete, depth of primer seating, primer crimp missing, loose primer, maximum profile and alignment and presence of foreign matter on the primer. The sample test cartridges shall be linked and fired at ambient temperature in bursts of 25 cartridges at a cyclic rate of not less than 650 nor greater than 1000 rounds per minute (RPM) in the M139 gun. The gun barrel shall be at ambient temperature at the beginning of the test and cooled to ambient temperature after every 50 rounds.

4.5.4.2 Function and casualty for primers for use in 25mm, cartridges and weapon systems. The method of test shall be as specified in AS12013566. The test sample primers shall be assembled into cartridge cases that have been previously accepted. In addition, each cartridge case shall be inspected for depth and diameter of primer pocket and length to shoulder basic diameter. The sample primed cases shall be inspected for absence of the booster. After assembly as a Cartridge, 25mm TP-T, M793, Dwg. 12013224, each test cartridge shall be inspected for projectile crimp improper or incomplete, depth of primer seating, primer crimp missing, loose primer, maximum profile and alignment and presence of foreign matter on the primer. The sample test cartridges shall be linked and fired at ambient temperature in the M242 gun in bursts of approximately 25 cartridges at a cyclic rate of not less than 175 nor greater than 225 RPM. The gun barrel shall be at ambient temperature at the beginning of the test and cooled to ambient temperature after every 100 rounds.

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4.5.5 Defective weapon. If the test weapon fails to comply with the cyclic rate established for the weapon or any test cartridge firing defect is found to have been caused by a defective weapon, as determined by the Government, the test shall be suspended. The weapon shall be corrected or replaced and the test repeated in whole or in part as indicated. If the test cartridge defect is not found to have been caused by a defective weapon it shall be valid for first article quantity or lot penalty.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level B. (See 5.2.1)

5.2 Packing.

5.2.1 Level B. The primers shall be packed in accordance with Dwg. 9387196.

5.3 Marking. Marking shall be in accordance with Dwg. 9387196.

5.4 Shipping. When primers from more than one lot are shipped at one time each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The primers covered by this specification are intended for use in 20mm and 25mm cartridges.

6.2 Ordering data. Ordering data should include:

a. Title, number and date of this specification.

b. Quantity required, delivery schedules, and cartridge and weapon system that primers are being procured for.

c. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

d. First Article Sample requirements, if other than specified in Section 4 of this specification.

e. Packaging requirements, if other that specified in Section 5 of this specification.

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f. Responsibility for furnishing test weapons, links, and parts and propellant to fabricate test cartridges.

g. Certificates of conformance for each lot or shipment of primers.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, ARDEC, ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001. Request letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test.

6.4 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as: Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture or an operational change which alters the metallurgical, chemical or physical properties of the item.

6.7 Definitions.

6.7.1 Hangfire. A hangfire is defined as any cartridge having an action time greater than four milliseconds in the M139 gun or six milliseconds in the M242 gun. Action time is defined as the time between application of the initiating energy to the primer and emergence of the projectile from the gun muzzle.

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